IT PROJECT MANAGEMENT · CHAPTER 08

Human Resource Management

The project team is the resource that delivers everything else. This chapter covers acquiring, organising, developing and leading the people on an IT project.

PMBOK Guide 7th ed. · Team performance domain **Scrum Guide 2020** **Case study: Realtime Quiz**

AGENDA

What we will cover

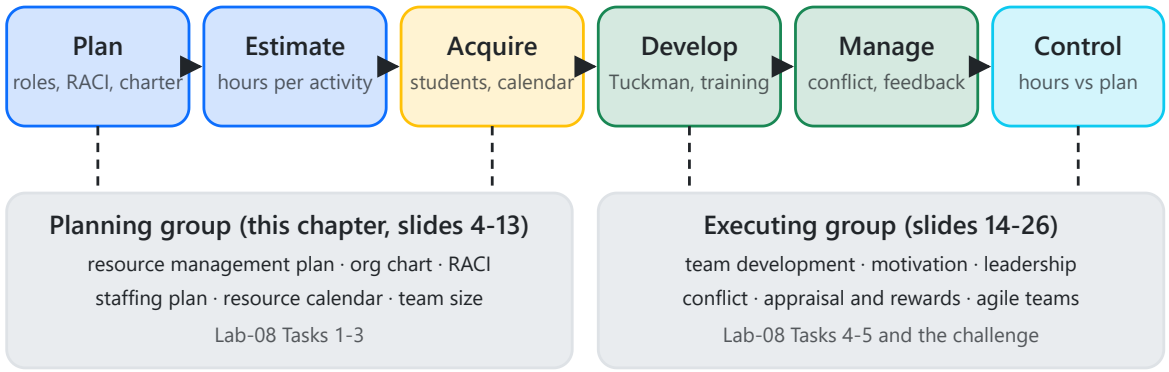
1. Resource management plan: roles, responsibilities, competencies
2. Organisation charts and responsibility assignment matrices (RACI)
3. Staffing management plan: acquisition, calendars, release, training
4. Estimating team size and the mythical man-month
5. Team development: Tuckman stages, ground rules, co-location and remote teams
6. Motivation theories applied to IT teams (Maslow, Herzberg, McGregor, self-determination)
7. Leadership styles, power and influence of the project manager
8. Conflict management techniques
9. Performance appraisal, recognition and rewards
10. Agile teams: self-organisation, cross-functional roles, servant leadership
11. Chapter Quiz
12. Lab-08

Click any item to jump directly to that topic.

People deliver the plan: six resource processes, one team

- Chapters 04 to 07 produced a scope, a schedule, a budget and a quality plan for the **Realtime Quiz** project. None of it happens without **people** who have time, skills and motivation.
- Our team: **4 students** (backend, frontend, QA/DevOps, product owner), a **supervising lecturer** as project manager and the **Head of the IT department** as sponsor. 14 weeks, pilot with 50 students in week 14.
- The PMBOK Guide 6th ed. named six processes (plan, estimate, acquire, develop, manage, control resources); the **PMBOK Guide 7th ed.** folds them into the *Team* performance domain: shared ownership, high-performing teams, leadership behaviours.
- The output of this chapter and of Lab-08 is the **resource management plan**: organisation chart, RACI, staffing plan, team charter.

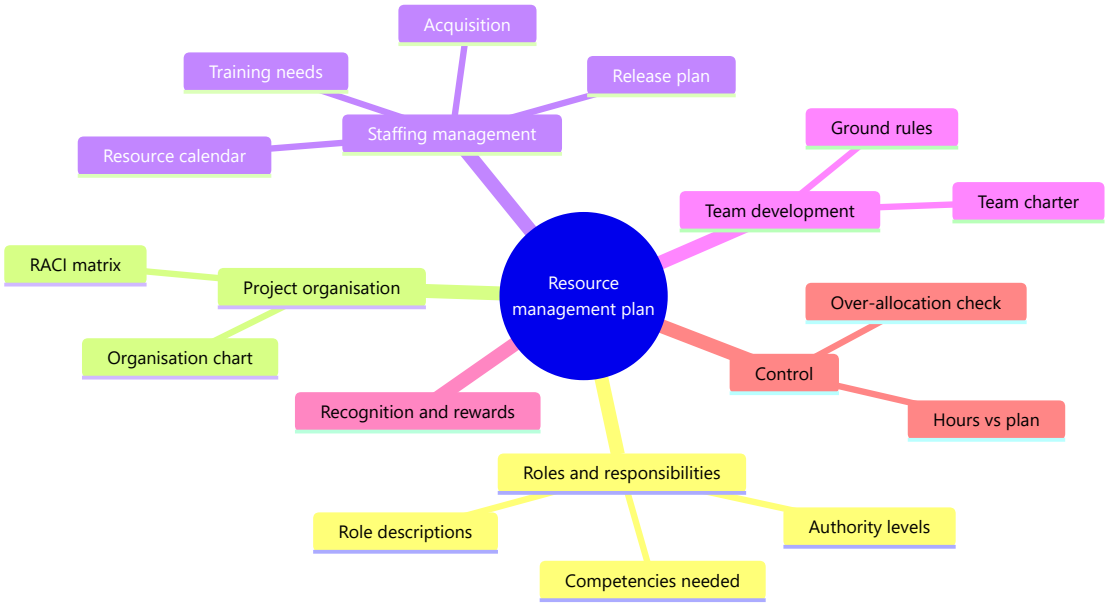
Resource management processes (PMBOK 6th) and what they produce here



TOPIC 1

The resource management plan: what it contains

- A subsidiary plan of the project management plan (PMBOK Guide 7th ed., *Planning* performance domain). It answers: **who** is on the project, **what** each person owns, **when** they are available and **how** they are developed and recognised.
- Four words to keep apart: a **role** is a hat (backend developer); a **responsibility** is work the role does (implement REST API); **authority** is what the role may decide (merge to `main`); a **competency** is a skill needed to do it (Spring Boot, SQL).
- Physical resources (Linode VPS, domain, laptops) belong to the same plan; this chapter concentrates on team resources.
- Keep it short: two pages plus the org chart, the RACI and the resource calendar. Everything else is a reference to Labs 04-06.



Worked example: a role description and the competency matrix

```
## Role: Backend developer (Dara)
Reports to: project manager (supervising lecturer)
Responsibilities
- Owns the Spring Boot code: web, realtime, service,
  domain, repository, security packages
- Designs the Flyway schema with the product owner
- Reviews every backend merge request within 2 days
Authority
- Merges to `main` after one approving review
- Decides library choices inside the pinned BOM
- Cannot change scope: proposes a change request
Competencies (target level 1-4, current level)
- Spring Boot 4, Spring Data JPA        target 3, now 2
- WebSocket / STOMP, concurrency        target 3, now 1
- PostgreSQL, SQL                       target 2, now 2
Time: 10 h/week, weeks 1-14, see resource calendar
Training: WebSocket lab (Ch. 1 of SE course) before W5
```

Competency (target)	Dara	Sokha	Vichea	Malis
Spring Boot, JPA (3)	2	1	1	1
WebSocket, STOMP (3)	1	1	0	0
HTML, Bootstrap, JS (3)	1	3	1	2
Testing, JUnit, load test (3)	2	1	2	0
Docker, Linux, CI (2)	1	0	2	0
Requirements, UX writing (2)	1	2	1	3

Levels: 0 none · 1 has seen it · 2 can do with help · 3 works alone · 4 can teach. A cell below its target is a **training need** (Task 3 of the lab) or a **risk** (Chapter 09).

The gap that matters most is the one on the critical path: WebSocket skills are needed in week 5 for the live session core, so the training must finish in week 4.

TOPIC 2

Organisation structures decide how much authority the PM has

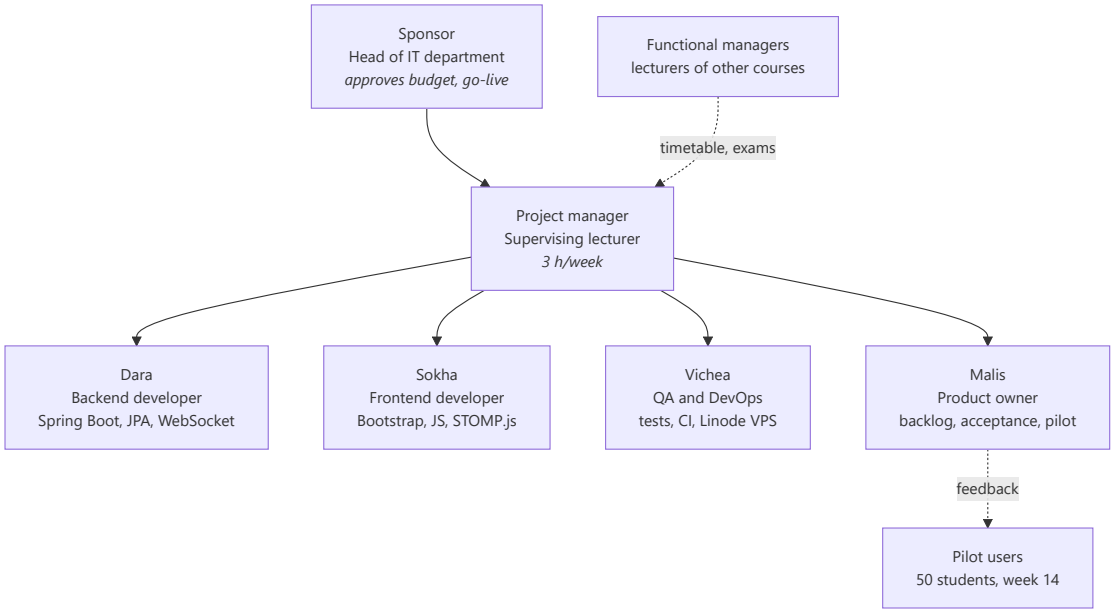
Structure	Who controls staff and budget	PM authority	Typical IT example
Functional	Department heads; staff borrowed part-time	Low: coordinator	An upgrade run by the IT department itself
Weak matrix	Functional managers; PM is an expediter	Low to limited	A student project inside a course timetable
Balanced matrix	Shared; PM and functional manager negotiate	Moderate	Software team lent to a product project
Strong matrix	PM has a PMO behind it; staff assigned mostly full-time	Moderate to high	Consulting delivery projects
Project-oriented	PM owns staff and budget for the project's duration	High	A dedicated product launch team

PMBOK Guide 7th ed. treats these as *organisational system* factors the PM must tailor to. Realtime Quiz runs in a **weak matrix**: students belong to the course timetable and to other courses; the lecturer influences through expertise and grades, not through a payroll.

- An **organisation chart** (OBS, organisational breakdown structure) shows reporting lines: who escalates to whom, who signs off. It is a picture of authority, not of the WBS.
- A **responsibility assignment matrix (RAM)** links WBS elements or activities to roles. The most used form is **RACI**: Responsible, Accountable, Consulted, Informed.
- In a weak matrix the org chart also names the functional managers the PM must negotiate with: here, the lecturers of the other courses whose exams fall in weeks 7 and 13.

⚠ A frequent lab mistake is to draw the WBS boxes into the org chart. WBS boxes are *deliverables*; org chart boxes are *people or roles*.

Worked example: the Realtime Quiz project organisation chart




```
flowchart TB
  S["Sponsor<br/>Head of IT department"]
  PM["Project manager<br/>Supervising lecturer"]
  S --> PM
  PM --> B["Dara<br/>Backend developer"]
  PM --> F["Sokha<br/>Frontend developer"]
  PM --> Q["Vichea<br/>QA and DevOps"]
  PM --> P["Malis<br/>Product owner"]
  P -.->|feedback| U["Pilot users<br/>50 students"]
  FM["Functional managers"] -.->|exams| PM
```

Solid lines are reporting lines (escalation, sign-off). Dotted lines are influence without authority. The functional managers box is what makes this a matrix: the PM does not control the students' calendars.

Worked example: RACI for the ten Lab-05 activities

Activity (Lab-05)	Sponsor	PM	Dara BE	Sokha FE	Vichea QA	Malis PO
A1 Setup, repo, CI, Docker Compose (W1)	I	C	R	I	A	I
A2 Accounts and JWT security (W2)	-	I	A/R	R	C	I
A3 Quiz authoring API and editor (W3-4)	-	I	R	R	C	A
A4 Live session core, WebSocket (W5-7)	-	I	A/R	R	C	C
A5 Scoring and leaderboard (W8)	-	I	A/R	C	R	I
A6 Analytics and CSV/PDF export (W9)	-	I	R	R	C	A
A7 Notifications and media (W10)	-	I	R	A/R	C	C
A8 Testing and hardening, load test (W11-12)	I	C	R	R	A/R	I
A9 Documentation, UML, final report (W13)	I	A	R	R	R	R
A10 Go-live decision and pilot (W14)	A	R	C	C	R	R

Responsible does the work · **A**ccountable answers for the result and signs it off (exactly one per row; A/R when the same person also does the work) · **C**onsulted gives input before · **I**nformed is told after.

Validation rules for a RACI

- Exactly **one A** per row. Two As means nobody decides; zero As means nobody answers for it.
- At least **one R** per row, otherwise the work is unassigned.
- No empty row and no empty column: a person with only I's is a stakeholder, not a team member.
- Read each **column**: too many R's on one person is an early sign of over-allocation (slide 10).
- Too many C's slows decisions; too few I's causes surprises.

✓ Column check above: Dara carries R or A on 9 of 10 rows. Task 3 of the lab tests whether the resource calendar can absorb that.

Staffing management plan: when people arrive, how much they can give, when they leave

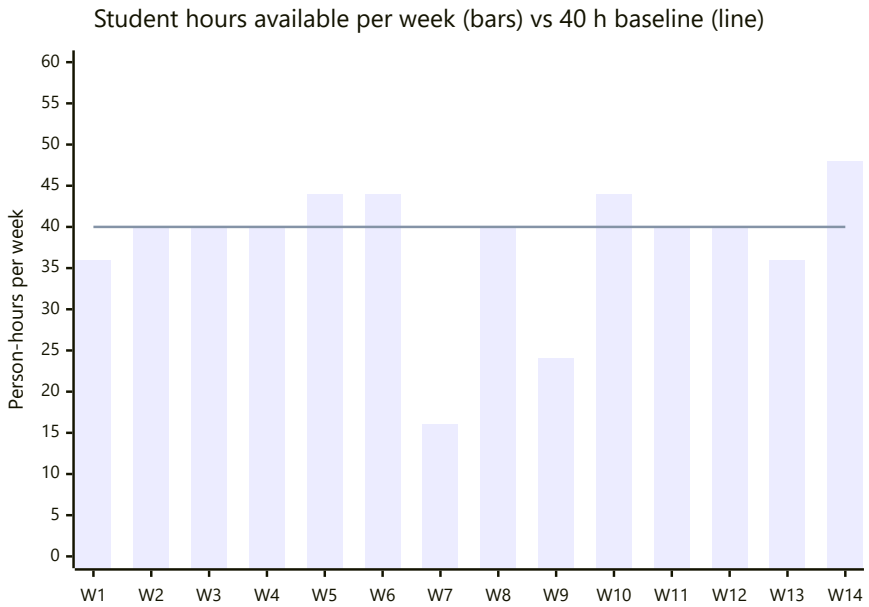
Part	Question it answers	Realtime Quiz
Acquisition	Where do people come from, pre-assigned or negotiated?	4 students pre-assigned by the course; lecturer negotiates lab room access
Resource calendar	Which hours can each person give per week?	10 h/week nominal, less in exam and holiday weeks
Training	Which competency gaps must close, by when?	WebSocket lab before W5; Docker before W1 for Vichea
Release	When does each person roll off, and how is knowledge kept?	All roll off after the pilot in W14; README and runbook by W13
Compliance	Rules on hours, safety, data	Max 12 h/week per student; pilot data anonymised

Worked example: resource calendar (available hours per person per week; W1 starts 2026-09-29). The calendar drives the histogram on the next slide and the levelling in ProjectLibre.

Week	Dara	Sokha	Vichea	Malis	Students	PM	Note
W1	9	9	9	9	36	3	kick-off, onboarding
W2-W4	10	10	10	10	40	3	normal weeks (x3)
W5-W6	11	11	11	11	44	3	live session core, peak
W7	4	4	4	4	16	2	midterm exams (all courses)
W8	10	10	10	10	40	3	normal week
W9	6	6	6	6	24	1	Water Festival, 3 days off
W10	11	11	11	11	44	3	catch-up week
W11-W12	10	10	10	10	40	3	testing and hardening (x2)
W13	9	9	9	9	36	3	final exam revision
W14	12	12	12	12	48	4	pilot with 50 students
Total	133	133	133	133	532	40	student hours = 532

⚠ The exam week W7 sits inside activity A4 (live session core, W5-7). Either the Lab-05 duration already used the calendar, or A4 needs 3 weeks of calendar time for 2 weeks of effort.

Staffing histogram, over-allocation and resource levelling



- A **resource histogram** plots hours needed (from the schedule) or hours available (from the calendar) per period. Overlaying both shows **over-allocation**: demand above the person's calendar.
- Worked check, Dara in W7: A4 needs 12 h of backend work that week, the calendar gives 4 h. **Over-allocated by 8 h.**
- **Resource levelling** (ProjectLibre: Tools, Level resources) delays non-critical work until capacity exists; it usually *extends* the schedule. **Resource smoothing** moves work only inside available float, so the end date holds.
- Other options: swap work between people (Sokha takes the STOMP client part), or reduce scope of that week (reconnect logic to W8).

```
Demand W7 (Dara)  = 12 h   Available W7 = 4 h
Over-allocation   = 12 - 4 = 8 h  -> level: move 8 h to W8
Available W8      = 10 h, planned 10 h -> W8 now 18 h: still over
Smooth instead    : 4 h W7, 6 h W6 (float), 2 h W8 -> fits
```

TOPIC 4

Estimating team size: effort, duration and headcount are not interchangeable

- **Effort** is work in person-hours (from Lab-06). **Duration** is calendar time. **Headcount** is the number of people. The naive formula `duration = effort / (people x hours per week)` holds only for work that can be split perfectly.
- Fred Brooks, *The Mythical Man-Month* (1975): a person-month is a unit of effort, not of progress. People and months are interchangeable only when a task is **partitionable** and needs **no communication**.
- Software work is partly sequential (design before code, deploy before test) and communication-heavy. Adding people adds **training time** and **communication channels**.
- Rule of thumb for student projects: keep the team at 3 to 6, and size the *work* to the team rather than the team to the work.

```
Naive check for the whole project
  Effort (Lab-06 estimate)      = 520 person-hours
  Team capacity per week        = 4 students x 10 h = 40 h
  Duration                      = 520 / 40 = 13 weeks   -> 1 week of slack
  Calendar reality (slide 9)    = 532 h available, 12 h buffer (2 %)

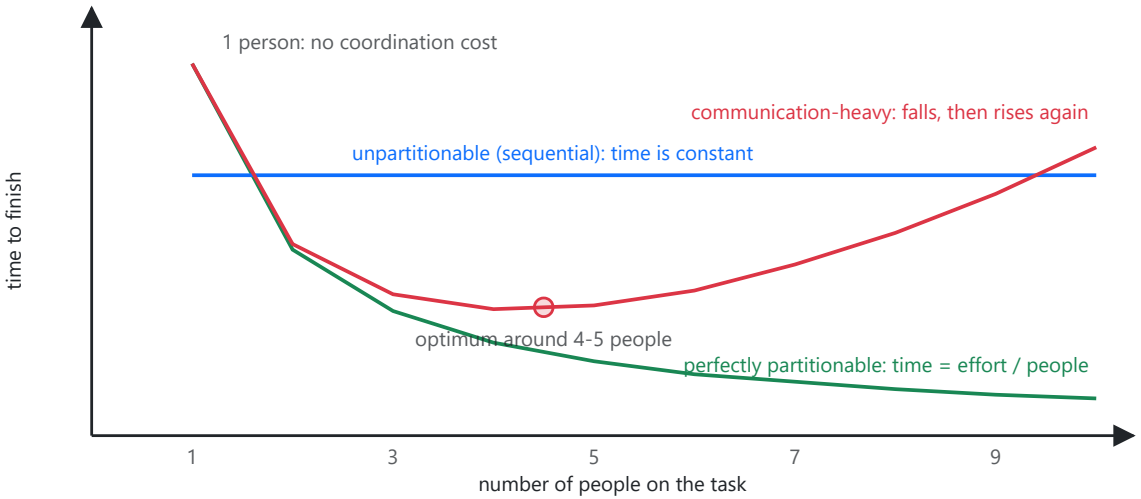
Communication channels  n(n-1)/2
  4 students + PM = 5 people    -> 5 x 4 / 2 = 10 channels
  add one student = 6 people    -> 6 x 5 / 2 = 15 channels (+50 %)
  add two         = 7 people    -> 7 x 6 / 2 = 21 channels (+110 %)

Sequential part: A4 live core -> A5 scoring -> A8 test -> A10 pilot
  cannot be shortened by headcount: each step needs the previous one
```

💡 The 2 % buffer is the real finding: the plan has no room for a sick week. Chapter 09 turns this into a risk with a response.

TOPIC 4

Brooks's law: "adding manpower to a late software project makes it later"



- **Why it holds.** A newcomer produces little at first, consumes the time of the people who train them, and adds communication channels for everyone. The work also has to be *re-partitioned*, which is planning work nobody budgeted.
- **When it does not hold.** Early in the project, for genuinely independent work (writing test data, documentation), or when the newcomer already knows the code base.
- **What to do instead when late:** cut scope (the Could list), extend the date, remove blockers, or pay down communication cost (fewer meetings, clearer interfaces).
- Related: the **Ringelmann effect**, individual effort falls as group size grows, and Amdahl-style reasoning: the sequential fraction caps the speed-up.

⚠ Brooks is about *late* and *software*. Doubling the people who paint a fence works fine.

Worked calculation: adding a fifth student in week 10

```
Situation at the start of W10 (A4 slipped by two weeks)
  Remaining effort: A5-A6 leftovers 50 h
                 + A7..A10 (40+70+45+45) 200 h  = 250 h
  Team capacity W10..W14 (44+40+40+36+48)     = 208 h
  Shortfall                                    =  42 h

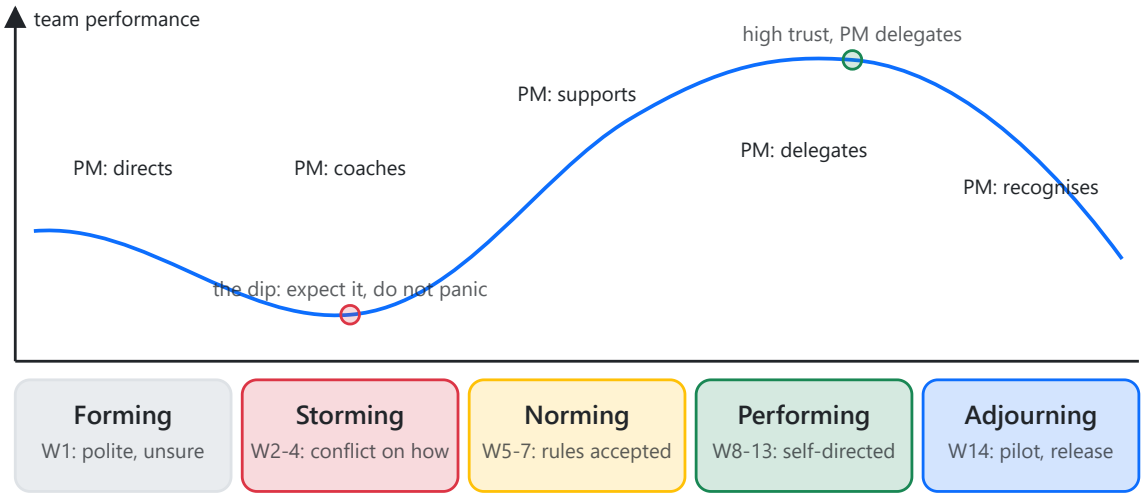
Option: add Piseth (5th student, 10 h/week, W10..W14)
  Gross hours                 5 weeks x 10 h    = +50 h
  Ramp-up: W10-W11 at 30 %    2 x 3 h effective =  +6 h  (not +20)
           W12-W14 at 80 %    3 x 8 h effective = +24 h  (not +30)
  Effective newcomer output                     = +30 h
  Mentoring by Dara            4 h x 2 weeks     =  -8 h
  Extra coordination: channels 10 -> 15,
           1 h/week x 4 existing x 5 weeks      = -20 h
  Re-partitioning work and reviewing            =  -6 h
  Net effect on the team                        =  -4 h

Result: shortfall grows from 42 h to 46 h. Adding a person is negative.
Alternatives: drop A7 media and chat (Could items)      -30 h of work
              move A9 report writing into W14 evenings  -12 h on path
              remaining gap 0 h, pilot stays in W14
```

- The calculation uses four costs Brooks named: **ramp-up**, **mentoring**, **communication** and **re-partitioning**. Only ramp-up is about the newcomer; the other three are paid by the existing team.
- The break-even moves with the horizon. Over 5 weeks the newcomer returns -4 h; over 15 weeks the same assumptions give +50 h. That is why hiring early is fine and hiring late is not.
- A late project should therefore **change scope or time**, not headcount. This is the triple constraint from Chapter 02 applied to people.
- In the lab challenge you repeat this calculation for the opposite case: a student *leaves* in week 8.

💡 When you must add people anyway: give them separable work (test data, docs, the load-test script), one mentor only, and a written onboarding checklist so the ramp-up is short.

Team development: the Tuckman ladder and the performance dip



- Bruce Tuckman (1965, adjourning added 1977): teams pass through **forming, storming, norming, performing** and **adjourning**. Stages can repeat when a member joins or leaves (the lab challenge).
- **Forming**: introduce roles, agree the charter, first small win (repo builds in W1). **Storming**: disagreements about tools and style; the PM keeps them on the *how*, not on persons.
- **Norming**: ground rules work without being cited. **Performing**: the team solves problems without the PM. **Adjourning**: hand-over, lessons learned, celebrate.
- The PM's style should **change with the stage** (slide 19): directing early, delegating late.

Team-building is any activity that moves the team one stage forward: a shared lunch counts, but a 30-minute retrospective every two weeks counts more.

Worked example: team charter and ground rules

```
# Team charter: Realtime Quiz (v1.0, agreed W1)
## Purpose
Ship a live quiz platform that 50 students use in W14.
## Values
Finish over perfect · say it early · no blame in retros
## Ground rules
1. Core hours Tue and Thu 14:00-17:00 in Lab B12.
2. Reply in the team chat within 24 h on weekdays.
3. Every change goes through a merge request; one
   review within 2 working days; author merges.
4. Definition of done = tests green + reviewed + demo.
5. Missing a core session: post the reason before it,
   and your update in the chat the same day.
6. Disagreements on technology: decide in 2 days with
   the PM as tie-breaker, then record in DECISIONS.md.
## Decision making
Product scope: Malis decides, PM can veto.
Technical: owner of the area decides after a review.
## Signatures
Dara · Sokha · Vichea · Malis · PM (lecturer)
```

- The **team charter** (PMBOK Guide 7th ed.) records values, communication rules, decision making and conflict handling. The team writes it, the PM facilitates, everyone signs. A charter imposed by the PM is a policy, not a charter.
- **Ground rules** are the testable part: each rule should let you say "this was broken" without argument. "Be respectful" is a value; "no interrupting in stand-ups" is a rule.
- Revisit the charter at the first retrospective (W3) when storming shows which rules are missing.

Co-located	Remote or hybrid
Osmotic communication: you hear the problem next to you	Everything must be written: chat, issues, decisions log
Whiteboard, pair programming on one screen	Shared draw.io board, screen-share pairing, more setup time
Fast trust, fast conflict	Slow trust; misread tone; schedule explicit social time
Fixed room and hours	Flexible hours, time zones; needs a cadence (next slide)

Worked example: communication cadence for a hybrid student team

Event	When	Who	Channel	Output
Async daily check-in	Weekdays by 09:30	4 students	Team chat thread	3 lines: done, next, blocked
Co-located work session	Tue and Thu 14:00-17:00	Students, PM drops in	Lab B12	Pairing, reviews, demos
Sprint review with PM	Friday 16:00, 30 min	Students + PM	Video call	Demo, hours logged vs plan
Retrospective	Every 2nd Friday after review	Students + PM	Video call, shared board	2 improvement actions, charter updates
One-on-one	Every 2 weeks, 15 min	PM with each student	In person or call	Feedback both ways, growth goal
Sponsor demo	End of W4, W8, W12	PM, Malis, sponsor	Sponsor's office	Go / no-go for next phase
Incident call	When CI red > 1 day or VPS down	Vichea + owner	Call within 2 h	Fix or rollback, note in chat

This table is the seed of the communication management plan in Chapter 10; here it is a **working agreement** for the days the team is not in the same room.

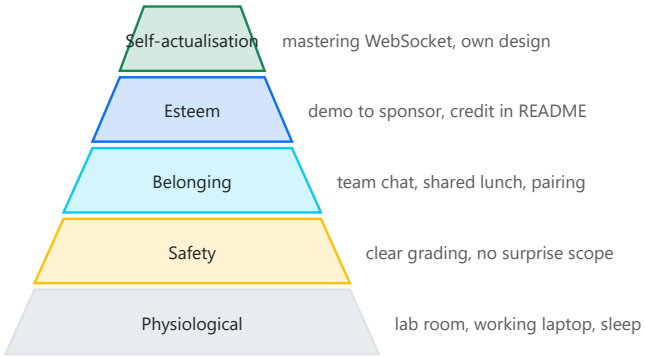
Working agreements for remote days

- Camera on for reviews and retros, optional for pairing.
- One topic per chat thread; decisions copied into `DECISIONS.md` the same day.
- "Blocked" in the check-in means someone replies within 2 hours during core hours.
- Merge requests carry a screenshot or a test log so a review needs no call.
- No messages expected after 21:00 or on holidays (W9 Water Festival).

⚠ Remote teams fail on *silence*, not on distance. The check-in exists so that a missing check-in is itself a signal the PM sees on the same day.

Motivation: Maslow's hierarchy and Herzberg's two factors

Maslow (1943): needs in order



a higher level motivates only once the lower ones are roughly met

Herzberg (1959): two factors

Hygiene absent: dissatisfied present: neutral	Motivators present: satisfied absent: neutral
working laptop	owning a module
fair grading rules	learning something new
VPS that stays up	public recognition
no weekend messages	seeing 50 users play
PM available	growing responsibility
clear roles	a real deployment

fix hygiene first, then invest in motivators
more hygiene does not create motivation

- **Maslow**: needs form a rough order; a student worried about failing another course (safety) will not be moved by "you can present at the demo" (esteem).
- **Herzberg**: *hygiene factors* (conditions, rules, pay, supervision) only prevent dissatisfaction. *Motivators* (achievement, recognition, the work itself, responsibility, growth) create satisfaction. They are two scales, not one.
- Practical reading for an IT PM: buying a faster laptop removes a complaint but does not motivate; letting Vichea own the deployment does.
- Both theories are old and criticised (no strict hierarchy, culture-dependent). Use them as checklists, not as laws.

McGregor's Theory X and Y, self-determination theory, and a motivation plan

Theory	Core idea	What the PM does
McGregor Theory X (1960)	People avoid work and need control and threats	Detailed task lists, hour logs checked daily, penalties. Works briefly, kills initiative
McGregor Theory Y	People seek responsibility when the work has meaning	Set goals and definition of done, let the team choose how; check outcomes weekly
Self-determination (Deci and Ryan): autonomy	Choice over how and when	Team picks tasks from the sprint backlog; core hours, not fixed hours
Self-determination: competence	Feeling effective and growing	Tasks slightly above current level, fast feedback from CI and reviews
Self-determination: relatedness	Belonging to people who care	Pairing, retros without blame, sponsor demo as a team
Extrinsic vs intrinsic	Grades and prizes can crowd out interest	Grade the process and the product; recognise learning, not only speed

Worked example: motivation plan (part of the resource management plan, revisited at each retrospective).

```
Person  Signal observed           Need      Action (owner, by)
Dara    stays late, silent in     autonomy  own the realtime package
        reviews                             end-to-end; PM stops
                                            micro-reviewing (PM, W3)
Sokha   proposes React, rejected  competence UI component library as
                                            her module; React spike
                                            after pilot (PM, W5)
Vichea  "my work is invisible"    esteem    deploy demo to sponsor in
                                            W8 is his slot (PM, W8)
Malis   overwhelmed by asks       safety    written backlog rules;
                                            PM handles sponsor asks
                                            (PM, W2)
Team    energy dip after exams    relatedness lunch after W7 exams;
                                            retro on wins (all, W8)
```

Leadership styles: match the style to the person and the situation

Style	How it looks	Fits when	Realtime Quiz moment
Directing (autocratic)	PM decides and tells; close follow-up	New team, crisis, safety or deadline at stake	W1 setup rules; VPS outage in W14
Coaching	PM explains why, asks questions, still decides	Willing but not yet able	Dara learning STOMP in W3-4
Supporting (participative)	Team decides, PM facilitates and removes blockers	Able but hesitant, storming	Frontend stack dispute W5
Delegating (laissez-faire)	PM sets outcome, stays out of the how	Able and committed; performing stage	Vichea owns the load test W11
Transactional	Clear goals, rewards for meeting them	Routine, measurable work	Hours logged and merged per sprint
Transformational	Vision, inspiration, individual attention	Change, uncertainty, need for energy	"50 students will use this" at kick-off
Servant	Leader serves the team: removes impediments, shields	Agile, self-organising teams (slide 25)	PM books the lab room, handles sponsor asks

The first four are the *situational leadership* quadrants (Hersey and Blanchard): choose by the follower's **competence** and **commitment**, and move towards delegating as both grow. PMBOK Guide 7th ed. lists laissez-faire, transactional, servant, transformational, charismatic and interactional styles.

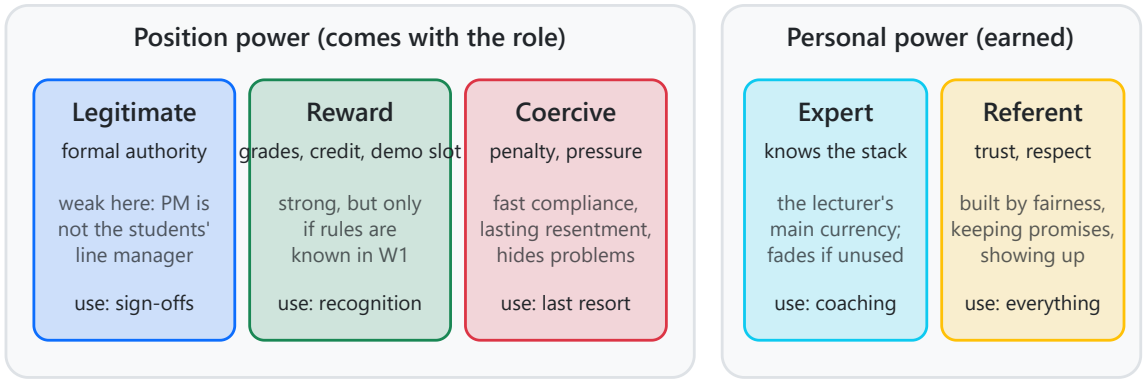
```
Situational grid (competence x commitment)

                    low commitment   high commitment
  low competence    COACHING         DIRECTING
                    (Sokha, W3       (whole team, W1
                     on JPA)          kick-off)
  high competence   SUPPORTING       DELEGATING
                    (Dara after       (Vichea on CI
                     A4 slipped)      and VPS, W11)

Style per Tuckman stage (slide 14)
  forming  -> directing      storming -> coaching
  norming  -> supporting     performing -> delegating
```

⚠ The common mistake is one style for everyone: delegating to a beginner is abandonment, directing an expert is insult.

Power and influence: what a PM in a weak matrix can actually use



French and Raven (1959). PMBOK Guide 6th ed. adds informational, situational, persuasive and pressure-based forms.

- **Influence without authority** is the normal state of an IT project manager in a matrix: the people report elsewhere. The PM's tools are expert and referent power, negotiation with functional managers, and the sponsor as escalation path.
- Order of preference for this project: **expert and referent** daily, **reward** through transparent grading and public recognition, **legitimate** for sign-offs, **coercive** almost never.
- Influence tactics that work with students: explain the *why*, ask for a small first commitment, make the work visible (demo), be predictable.

💡 Test for your own power base: if the grade did not exist, would the team still follow your advice? If not, invest in expert and referent power.

Conflict is normal: sources and the five techniques, ranked

#	Technique (PMBOK Guide 6th ed.)	What happens	Outcome	Use when
1	**Collaborate / problem-solve**	Open dialogue, all viewpoints, decide on criteria	Win-win, durable	Time exists, relationship and result both matter
2	**Compromise / reconcile**	Each side gives something	Lose-lose but acceptable	Equal power, time pressure, temporary fix
3	**Smooth / accommodate**	Stress agreement, play down difference	Relationship kept, issue stays	Stakes low, keep peace before a bigger talk
4	**Force / direct**	One side imposes, often via authority	Win-lose, fast	Emergency, safety, deadline in hours
5	**Withdraw / avoid**	Postpone or step back	No resolution	Cooling off needed, or issue will vanish by itself

Ranking is by how often the result lasts. Every technique is right somewhere; the failure is using the same one every time. Address conflict **early**, **in private**, on the **issue not the person**, and with a **direct, collaborative** approach first.

Where conflict comes from on IT projects

- **Schedules** and priorities: the most frequent source; who works on what first.
- **Scarce resources**: one lab room, one VPS, one reviewer.
- **Technical opinions**: framework, style, architecture. Healthy if about criteria, toxic if about ego.
- **Roles** unclear: two people think they own the same module (fix: RACI).
- **Personal style**: rare as a root cause, common as the visible symptom.

💡 Most conflicts are prevented by planning artefacts you already made: a RACI, ground rules, a definition of done and a decisions log.

Worked example: one conflict resolved five ways

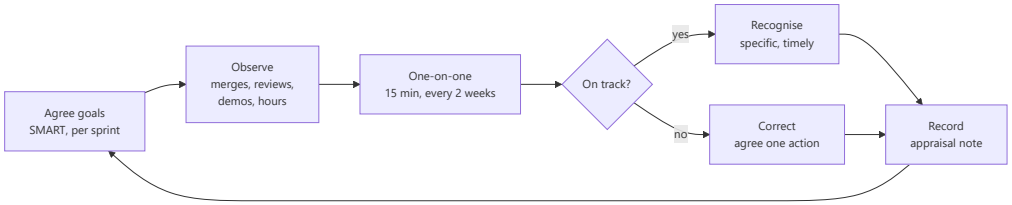
⚡ Scenario (week 5)

Sokha wants to rebuild the frontend in React: "the play page will be a mess in plain JS". Dara and Vichea want to keep Bootstrap and vanilla JS as planned: "no time, and the course stack is fixed". The lobby page has been stalled for three days; the chat thread has 40 messages and is getting personal. A4 (live session core) is on the critical path.

- **Recommended: collaborate.** The dispute is about criteria, both sides have a point, and the result must last 9 more weeks. Cost: one 45-minute meeting.
- Force would also unblock A4 today; use it only if the meeting cannot happen this week.
- Whatever the technique, record the decision in `DECISIONS.md` and revisit it at the retro, not in the chat.

Technique	What the PM does	Likely result
Withdraw	Says nothing, hopes it settles	A4 slips another week; resentment
Smooth	"Both good ideas, let's stay friends, back to work"	Stall continues, flares up in W7
Compromise	"React for the play page only, plain JS elsewhere"	Two stacks, two skill sets, more testing: worse
Force	"The plan says Bootstrap. Decision made. Next topic."	Unblocked today; Sokha disengages
Collaborate	45-min meeting, PM facilitates:	Plain JS now, one shared component library owned by Sokha; React spike in the Could list after the pilot; A4 resumes
1 goal	pilot in W14 with 50 students	
2 criteria	learning curve, time, course stack, maintainability	
3 options	React now / plain JS / component library	
4 score	each option against each criterion	
5 decide	record in DECISIONS.md, retro in W6	

Performance appraisal: a short feedback loop, not a yearly form



- **Appraisal** answers two questions: is the work meeting the plan, and is the person growing? Both need evidence: merged work, review quality, demo outcomes, reliability of hours, not impressions.
- **360-degree feedback** adds the view of peers and the product owner, useful in a team where the PM sees only 3 hours a week. Keep it structured: two strengths, one growth point per person.
- Feedback rules: **timely** (this week, not at the end), **specific** (the reconnect fix, not "good job"), **behaviour not person**, **private for correction, public for praise**.
- Recognition and rewards must match the culture and the resource plan: what is *promised* in W1 must be *deliverable* in W14. Do not reward overtime; you get more overtime.

```
One-on-one agenda (15 min)  1 Your week: wins, blockers
                            2 Feedback to you (1 specific item)
                            3 Feedback to me / the process
                            4 Growth goal check (from W1)
                            5 One action, written down
```

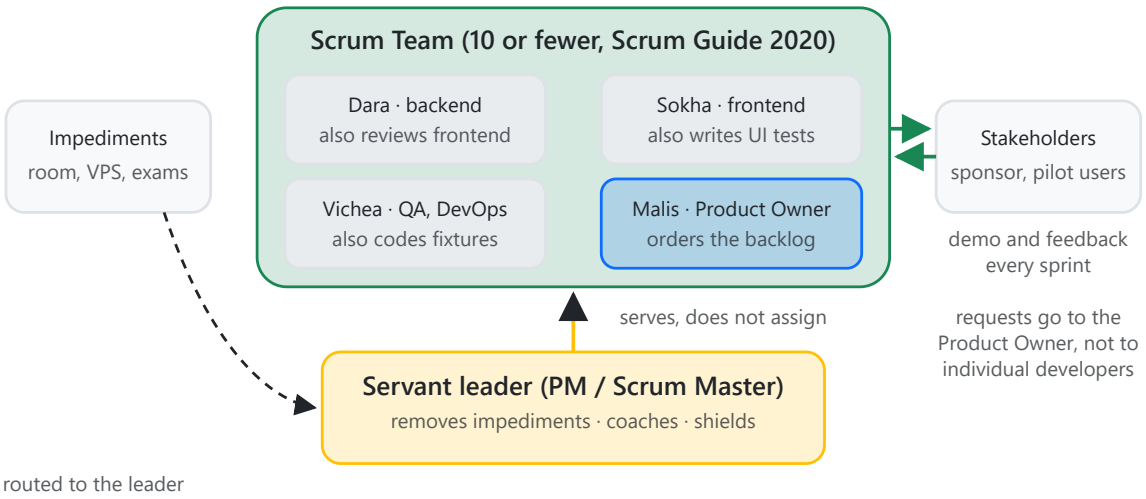
Worked example: one-on-one and recognition plan for the Realtime Quiz team

Element	Rule	Example	Cost
One-on-one	Every 2 weeks, 15 min, PM with each student, agenda from slide 23	W2, W4, W6, W8, W10, W12 plus a closing talk in W14	7 x 4 x 15 min = 7 h of PM time
Weekly shout-out	Friday review: one specific recognition per week, rotating	"Vichea's CI cache cut the build from 9 to 3 minutes"	0
Milestone credit	Name the owner in the sponsor demo and in the README	"Live session core by Dara and Sokha" (W8 demo)	0
Growth reward	Learning goal met: lead the next design talk	Sokha presents the component library at W10 review	30 min of review time
Team reward	Team-level, not individual, when a phase closes	Lunch after MVP checkpoint (W8), paid from the small cash budget	about 20 USD
Grading transparency	Grading rubric published in W1, individual and team parts	60 % team deliverable, 40 % individual contribution evidence	0
Correction path	Private, specific, one action, follow-up date	Missed check-ins: talk in W3, action: post by 09:30, check W4	15 min

- Recognition is **cheap and frequent**; rewards are **rare and planned**. The plan above spends under 8 hours of PM time and 20 USD over 14 weeks.
- **Individual evidence** for grading comes from the same sources as appraisal: merge history, review comments, demo ownership, one-on-one notes. Tell the students this in W1 (hygiene factor).
- Avoid **zero-sum** rewards ("best contributor") in a 4-person team: they punish collaboration, the thing you most need.
- Rewards must be reachable by everyone: a reward only the strongest can win demotivates the other three.

⚠ The classic failure: recognising the hero who fixed the outage at 02:00 and forgetting the person whose tests prevented three outages.

Agile teams: cross-functional, self-managing, served by a leader



- **Scrum Guide 2020**: one Scrum Team of 10 or fewer, no sub-teams or hierarchies; a Product Owner, a Scrum Master and Developers. The team is **cross-functional** (all skills needed to deliver) and **self-managing** (decides who does what, when, how).
- **Self-organisation** is not "no management": the goal, the sprint length, the definition of done and the boundaries are fixed; the path inside them is the team's.
- **Servant leadership** (Greenleaf; PMBOK Guide 7th ed.): the leader's success is measured by the team's growth and by impediments removed, not by decisions made.
- Cross-functional does not mean everyone does everything: it means the *team* has every skill and no task waits for an outsider.

TOPIC 10

Traditional PM roles and agile roles side by side, and how this team runs

Concern	Traditional (Ch. 08 so far)	Agile (Scrum Guide 2020)
Who assigns work	PM via RACI and schedule	Developers pull from the sprint backlog
Who owns scope	Sponsor via change control	Product Owner orders the backlog
Who removes blockers	PM escalates	Scrum Master serves the team
Team shape	Specialists, possibly part-time	Cross-functional, stable, dedicated
Appraisal	PM appraises individuals	Team inspects itself in the retrospective
Org chart	Hierarchy of roles	One team box plus stakeholders
Staffing plan	Calendar, histogram, levelling	Capacity per sprint in hours or points

Realtime Quiz is **hybrid** (Chapter 03): a fixed 14-week plan and budget with two-week sprints inside it. The RACI names owners for accountability; inside a sprint the developers still pull tasks.

```
Sprint capacity from the resource calendar (slide 9)
  Sprint 3 = W5 + W6   : 44 + 44 = 88 student hours
  Sprint 4 = W7 + W8   : 16 + 40 = 56 student hours  (exams)
  Sprint 5 = W9 + W10  : 24 + 44 = 68 student hours  (holiday)
  Reserve for reviews, retro, ops   : 15 % of capacity
  Sprint 4 planning budget          : 56 x 0.85 = 47.6 -> 47 h

Self-organisation rules agreed in the charter
  - anyone may take any backlog item; the area owner reviews it
  - a task blocked for a day is raised to the servant leader
  - no one holds more than 2 items in progress
  - the retro decides process changes, not the PM alone
```

Same numbers, two views: the histogram of slide 10 is the sprint capacity of an agile team, read two weeks at a time.

WRAP-UP

Best practices and common mistakes

1 Write roles before names

Define the role, its responsibilities, authority and competencies first; then assign a person. It makes replacement possible (the lab challenge) and keeps the RACI stable.

2 One A per row, then read the columns

The row check finds decision gaps; the column check finds the person who will burn out. Both take two minutes.

3 Put the calendar in the schedule

Exams and holidays are known in week 1. A schedule that ignores them is fiction; ProjectLibre calendars exist for this.

4 Fix scope or time, not headcount, when late

Brooks's law: a late newcomer costs ramp-up, mentoring and communication before returning anything. Run the calculation from slide 13 before asking for help.

5 Expect the storm

Tuckman's dip in weeks 2-4 is normal. Have the charter, a retrospective and a decisions log ready before it, not after.

6 Hygiene first, motivators second

Remove the complaints (room, laptop, unclear grading), then invest in ownership, learning and recognition. Neither substitutes for the other.

✕ Mistake: one leadership style

Directing experts or delegating to beginners. Match style to competence and commitment, and change it as the team matures.

✕ Mistake: avoiding the conflict

Withdrawing feels kind and costs a week. Collaborate early, in private, on the issue; force only in emergencies.

✕ Mistake: rewarding heroics

Praising the 02:00 fix teaches the team to create emergencies. Recognise the quiet prevention work and never reward overtime.

Chapter quiz 10 questions

1. In a RACI matrix, how many "A" (Accountable) entries should each activity row contain?

- As many as there are team members
- At least two, for redundancy
- Exactly one
- None; A is optional

2. The team grows from 5 people (4 students + PM) to 7. How many communication channels are there afterwards, using n(n-1)/2?

- 10
- 15
- 21
- 42

3. Remaining effort is 240 person-hours. Four students each give 10 h/week and the work is perfectly partitionable. What is the duration?

- 4 weeks
- 6 weeks
- 8 weeks
- 24 weeks

4. A newcomer joins for 5 weeks: 30 h of effective output, 8 h of mentoring lost, 20 h of extra coordination, 6 h of re-partitioning. Net effect on the team?

- +30 h
- +4 h
- -4 h
- -34 h

5. Which statement is Brooks's law?

- Work expands to fill the time available
- Adding manpower to a late software project makes it later
- Any team will storm before it performs
- Ninety percent of a project takes ninety percent of the time

6. In week 3 the team argues about coding style and tool choices, and productivity dips. Which Tuckman stage is this?

- Forming
- Storming
- Norming
- Performing

7. According to Herzberg, which of these is a hygiene factor rather than a motivator?

- Owning the realtime module end-to-end
- Public recognition at the sponsor demo
- A lab room that is available at the agreed hours
- Learning WebSocket programming

8. Two developers disagree on the frontend framework; the critical path is stalled, both have valid points and the decision must last for nine weeks. Which technique is recommended first?

- Withdraw and let it cool down
- Force the planned choice
- Collaborate: criteria, options, decide together
- Compromise: use both frameworks

9. A student has 4 available hours in the exam week but the schedule assigns 12 h of work. What is this called and what technique keeps the end date?

- Over-allocation of 8 h; resource smoothing inside float
- Under-allocation of 8 h; crashing
- Over-allocation of 8 h; resource levelling always keeps the end date
- A scope gap; fast-tracking

10. Which style should a PM use with a team member who is highly competent and highly committed (situational leadership)?

- Directing
- Coaching
- Supporting
- Delegating

WRAP-UP

Summary

The resource management plan defines roles, responsibilities, authority and competencies before it names people.

The org chart shows authority; the RACI links activities to roles with exactly one A per row and at least one R.

The staffing plan covers acquisition, a resource calendar with exams and holidays, training needs and release.

Effort, duration and headcount are different things; Brooks's law makes a late newcomer a net cost (ramp-up, mentoring, channels n(n-1)/2).

Teams form, storm, norm, perform and adjourn; a charter, ground rules and a cadence carry them through the storm.

Hygiene factors prevent dissatisfaction; motivators (ownership, growth, recognition) create it. Theory Y and self-determination guide the PM.

Match leadership style to competence and commitment; in a weak matrix the PM lives on expert and referent power.

Five conflict techniques; collaborate first, force only in emergencies, never default to avoidance.

Appraise with evidence and short loops; recognise often, reward rarely; agile teams self-manage inside fixed goals with a servant leader.

Open Lab-08: 5 tasks + 1 challenge

Next chapter: 09 · Risk Management